

Fundamentals of Computer Science: Core Terminology

Educational Series

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1 The Computer System

A computer is an electronic device that processes data, converting it into information. It operates under the **IPO Cycle** (Input-Process-Output).

2 Hardware Components

Hardware refers to the physical, tangible parts of the computer.

- **CPU (Central Processing Unit):** The "Brain" of the computer. It consists of the *ALU (Arithmetic Logic Unit)* for calculations and the *CU (Control Unit)* for directing operations.
- **Motherboard:** The main circuit board that connects all components.
- **Input Devices:** Tools used to enter data (e.g., Keyboard, Mouse, Scanner, MICR).
- **Output Devices:** Tools that display or print results (e.g., Monitor, Printer, Plotter, Speakers).
- **Storage Devices:**
 - **Primary Memory:** *RAM* (Random Access Memory - Volatile) and *ROM* (Read Only Memory - Non-volatile).
 - **Secondary Memory:** *HDD* (Hard Disk Drive), *SSD* (Solid State Drive), and Flash drives.

3 Software Classifications

Software is a set of instructions (programs) that tells the hardware what to do.

1. **System Software:** Programs that manage the computer hardware (e.g., Operating Systems like Windows/Linux, Device Drivers).
2. **Application Software:** Programs designed for specific user tasks (e.g., MS Word, Web Browsers, Games).
3. **Utility Software:** Programs used for maintenance (e.g., Antivirus, Disk Defragmenter).

4 Key Terminology

- **Bit:** The smallest unit of data (0 or 1).
- **Byte:** A group of 8 bits.
- **Bootng:** The process of starting or restarting a computer.
- **GUI (Graphical User Interface):** A visual way of interacting with a computer using items like windows, icons, and menus.
- **BIOS (Basic Input/Output System):** Firmware used to perform hardware initialization during the booting process.
- **Cache Memory:** A small, high-speed memory area used by the CPU to reduce the average time to access data from the main memory.

5 Data Measurement Units

Unit	Equivalent
4 Bits	1 Nibble
8 Bits	1 Byte
1024 Bytes	1 KB (Kilobyte)
1024 KB	1 MB (Megabyte)
1024 MB	1 GB (Gigabyte)
1024 GB	1 TB (Terabyte)

6 Common Abbreviations

- **ALU:** Arithmetic Logic Unit
- **HTTP:** Hypertext Transfer Protocol
- **URL:** Uniform Resource Locator
- **PDF:** Portable Document Format
- **LAN/WAN:** Local Area Network / Wide Area Network